

Supermarket Sales Analysis

Retail | 2023

Total Revenue	Gross Income	Transactions	Avg Transaction	Gross Margin	Avg Rating
519,281	11,025	2000	259.64	2.1%	6.96 / 10

Executive Summary

This report analyses 2,000 real supermarket transactions across three city branches in Myanmar — Yangon, Mandalay, and Naypyitaw. The analysis covers revenue trends, product line performance, customer demographics, payment preferences, and peak shopping patterns. Customer ratings are on a 4 to 10 scale. Findings support inventory management, staffing decisions, and promotional planning.

Key Insights

- Health & Beauty is the top product line with revenue of 96,970.
- Yangon is the highest-revenue city among the three.
- Branch B outperforms the other two branches.
- Ewallet is the most preferred payment method.
- Peak sales hour is 18:00, the busiest window of the day.
- Average customer rating across all product lines is 6.96 out of 10.

Recommendations

- Increase Health & Beauty stock levels — it is the best-selling product line.
- Study Branch B practices and replicate them across lower-performing branches.
- Deploy extra staff at 18:00 to handle the peak sales window.
- Offer incentives to Normal customers to encourage membership conversion.
- Promote the least used payment method with cashback rewards.
- Concentrate campaigns in Yangon and extend successful tactics city-wide.

Figure 1

Daily Revenue Trend

This chart plots total daily sales across all three branches. Revenue fluctuates with customer traffic — weekends and mid-month periods tend to generate higher activity. The shaded area emphasises volume and makes quiet periods easy to identify. The chart covers the full dataset period and provides a high-level view of overall trading momentum.

Total Revenue	Peak Day	Daily Average
519,281	4,100	1,427

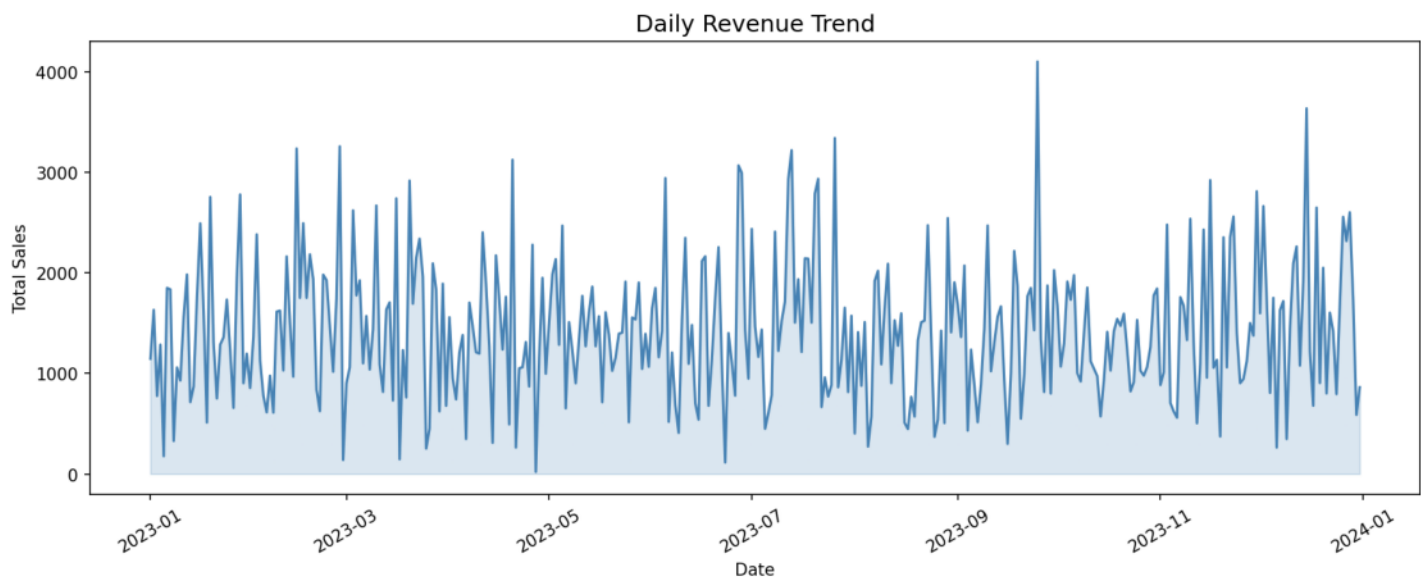


Figure 2

Total Sales by Product Line

The chart compares cumulative revenue across the six product lines. Health & Beauty leads overall. The relatively even spread across lines means no category is severely underperforming; however, top performers deserve priority in stock planning and promotional focus. Comparing revenue here against ratings in Figure 6 reveals whether high-selling lines also satisfy customers.

Health & Beauty	Sports & Travel	Electronic Accessories	Home & Lifestyle	Food & Beverages	Fashion Accessories
96,970	91,178	88,995	86,352	78,234	77,552

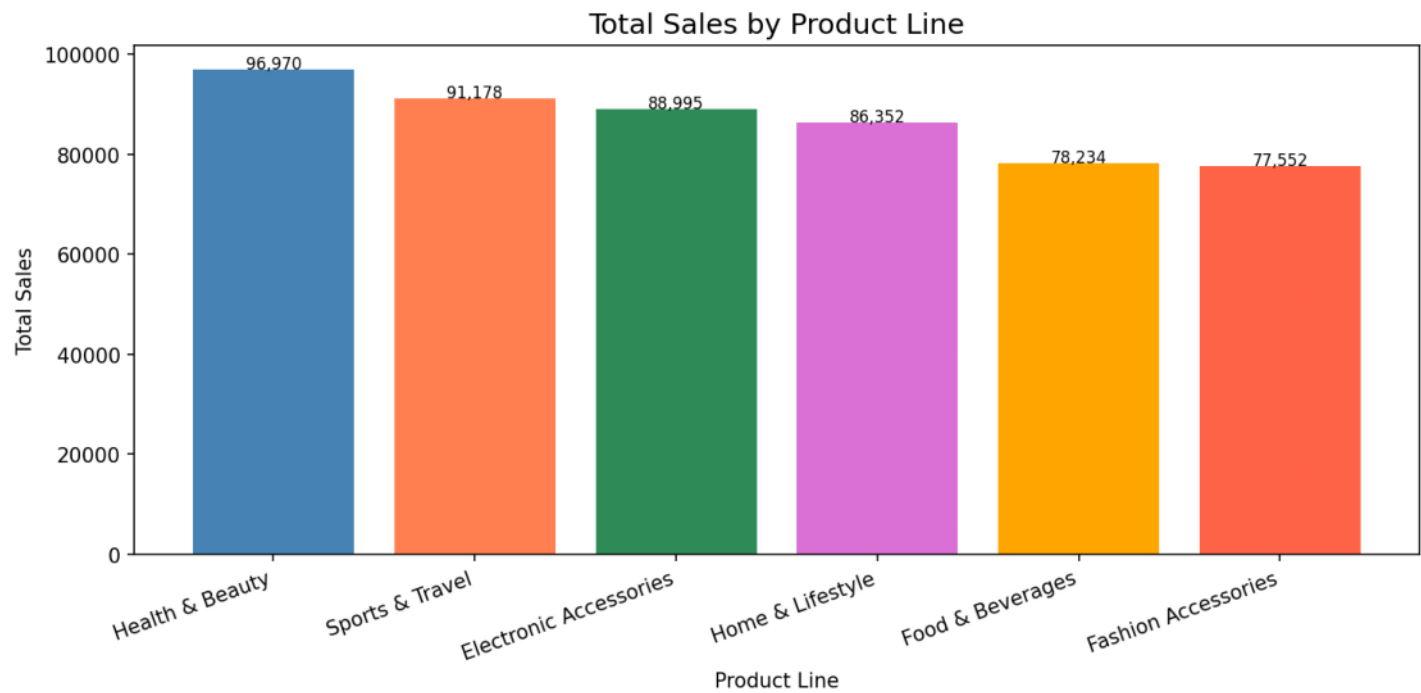


Figure 3

Revenue by Payment Method

Total revenue broken down by payment method. Ewallet leads but the distribution is broadly balanced across Cash, Credit card, and Ewallet. Tracking this split over time reveals whether digital payment adoption is growing, which affects checkout infrastructure investment decisions.

Most Used	Methods	Transactions
Ewallet	3	2000

Revenue by Payment Method

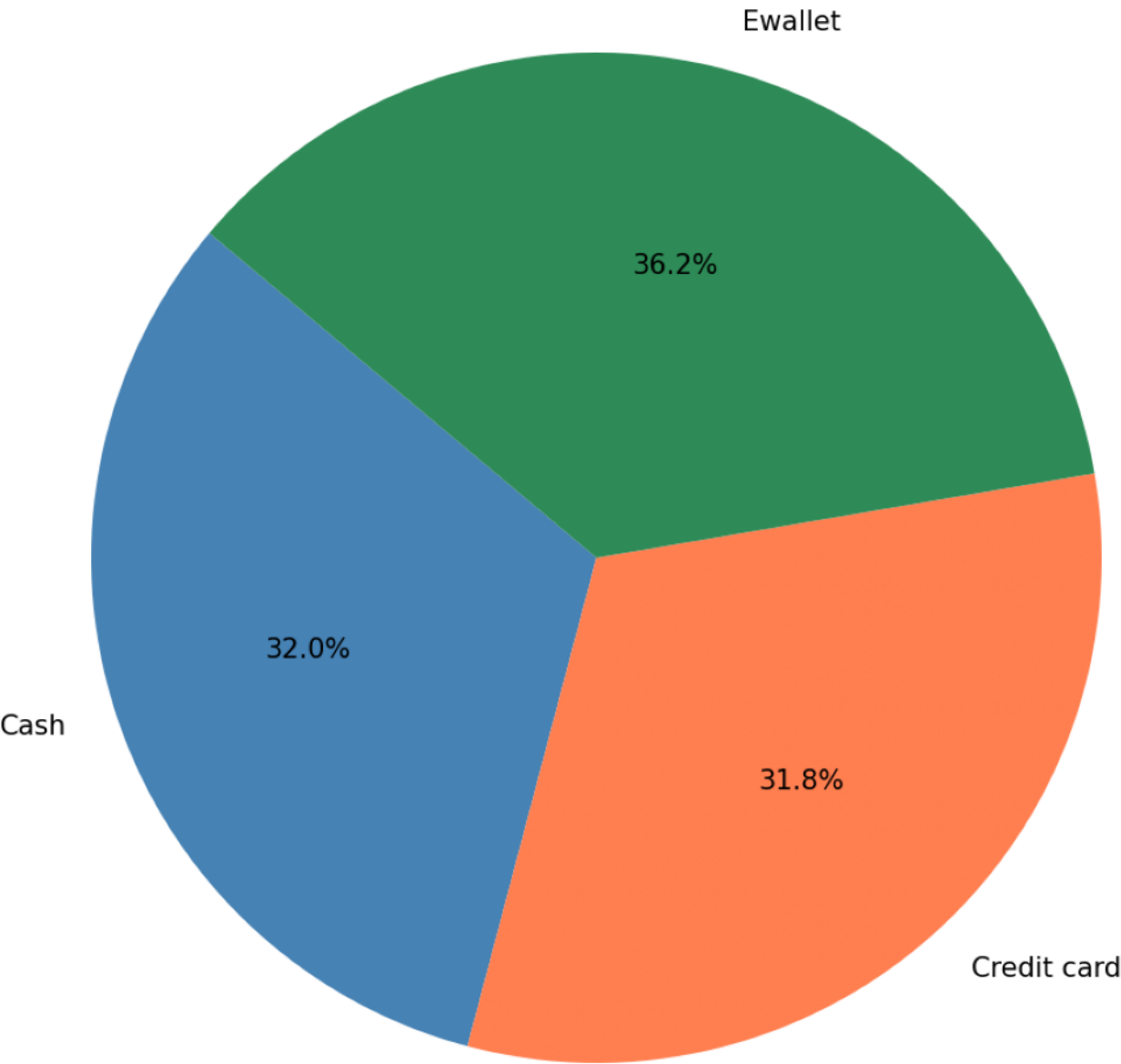


Figure 4

Sales by Branch and City

Two bar charts compare revenue at branch level (A, B, C) and city level. Branch B and city Yangon lead their respective comparisons. Gaps between branches point to differences in footfall, product mix, or operational efficiency. Understanding what the top branch does differently informs targeted improvements in others.

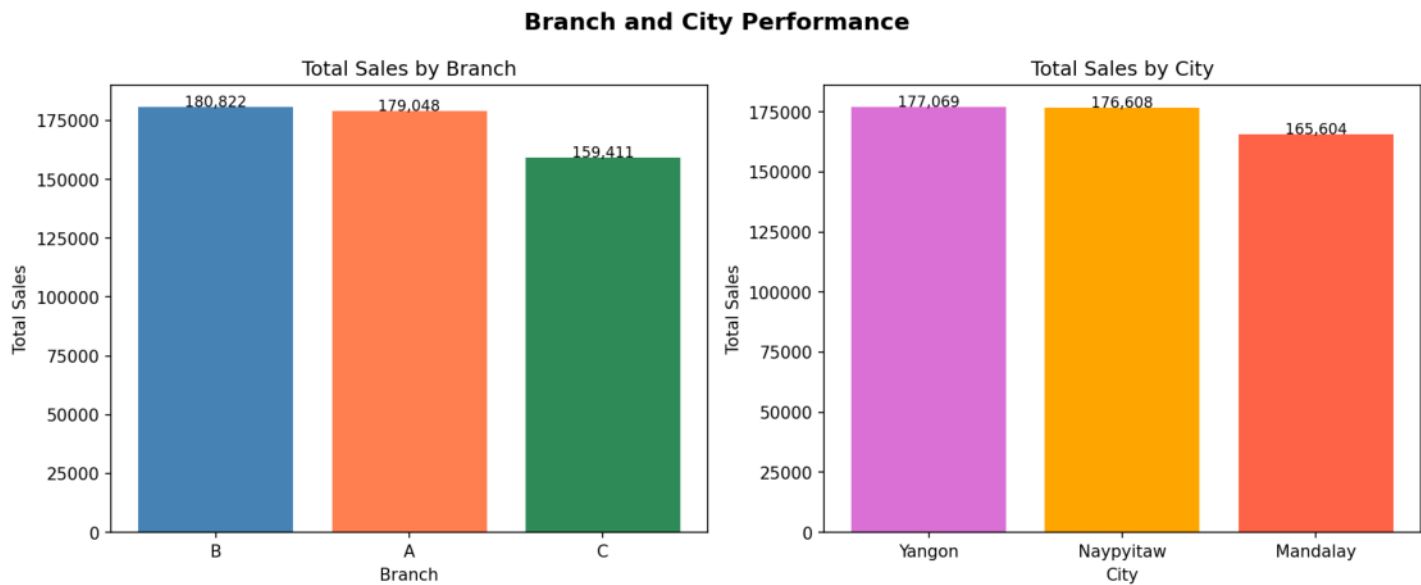


Figure 5

Customer Demographics

Revenue split by membership type and gender. Members enrolled in the loyalty programme tend to visit more frequently. The gender split identifies which product lines or promotions resonate with each group, supporting more targeted marketing and loyalty programme design.



Customer Demographics

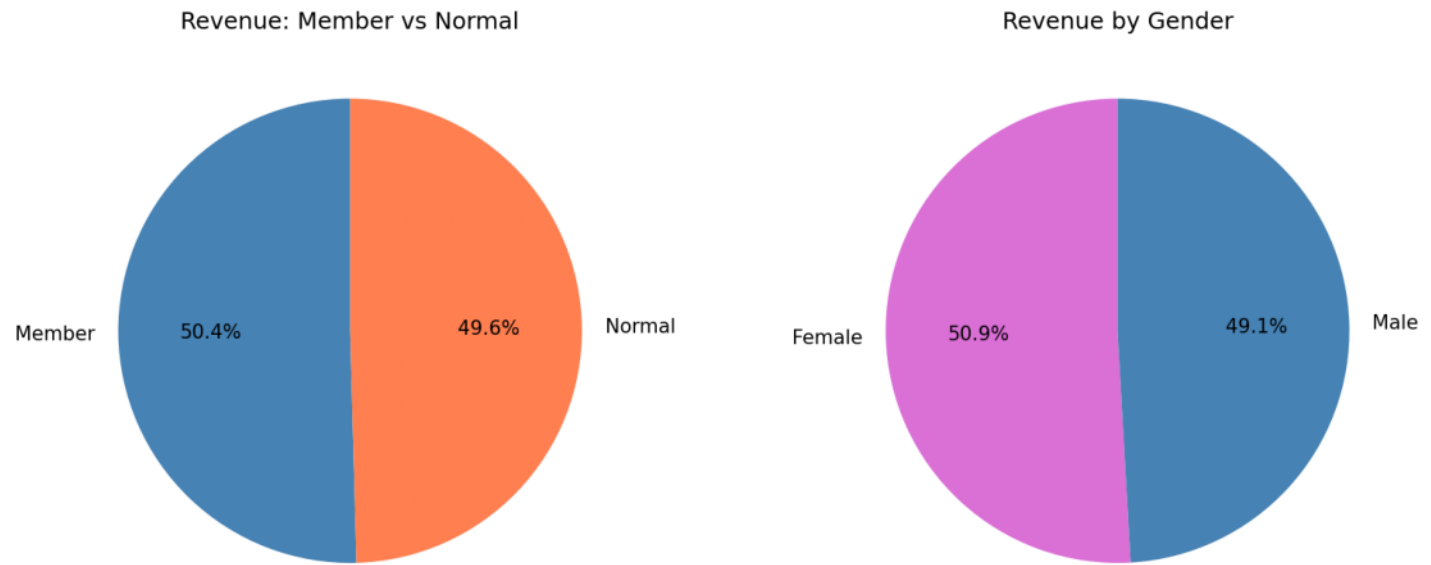


Figure 6

Average Customer Rating by Product Line

Average rating out of 10 per product line. Green bars exceed the overall average of 6.96; coral bars fall below it. This separates sales performance from customer satisfaction — a top-selling line with a below-average rating signals a quality gap that, if addressed, could further strengthen its position.

Overall Average	Highest Rated	Lowest Rated
6.96 / 10	Food & Beverages (7.05)	Electronic Accessories (6.86)

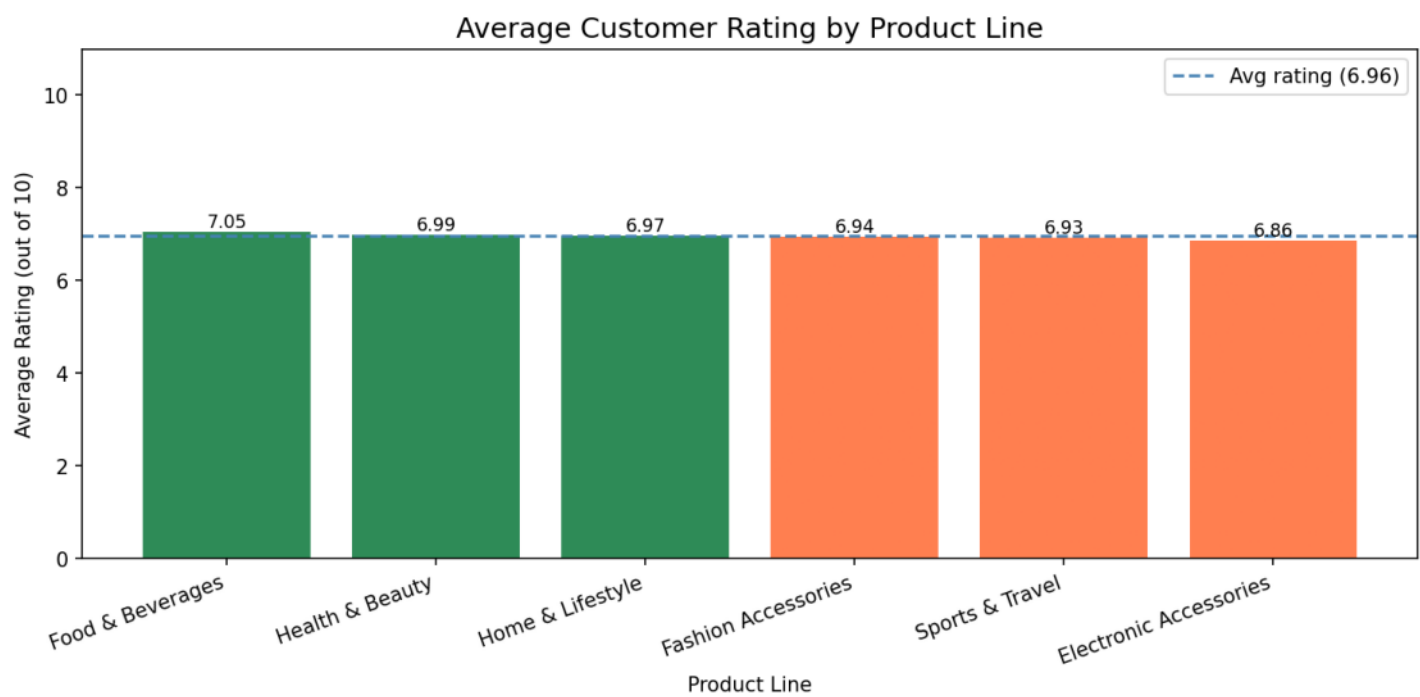


Figure 7

Sales by Hour of Day

Total revenue per hour across all branches. The peak hour is 18:00. Understanding the hourly distribution is critical for staff scheduling, restocking during quiet periods, and timing promotional messages. Hours with very low sales may be candidates for reduced operating hours.

Peak Hour	Quietest Hour	Peak Revenue
18:00	11:00	48,287

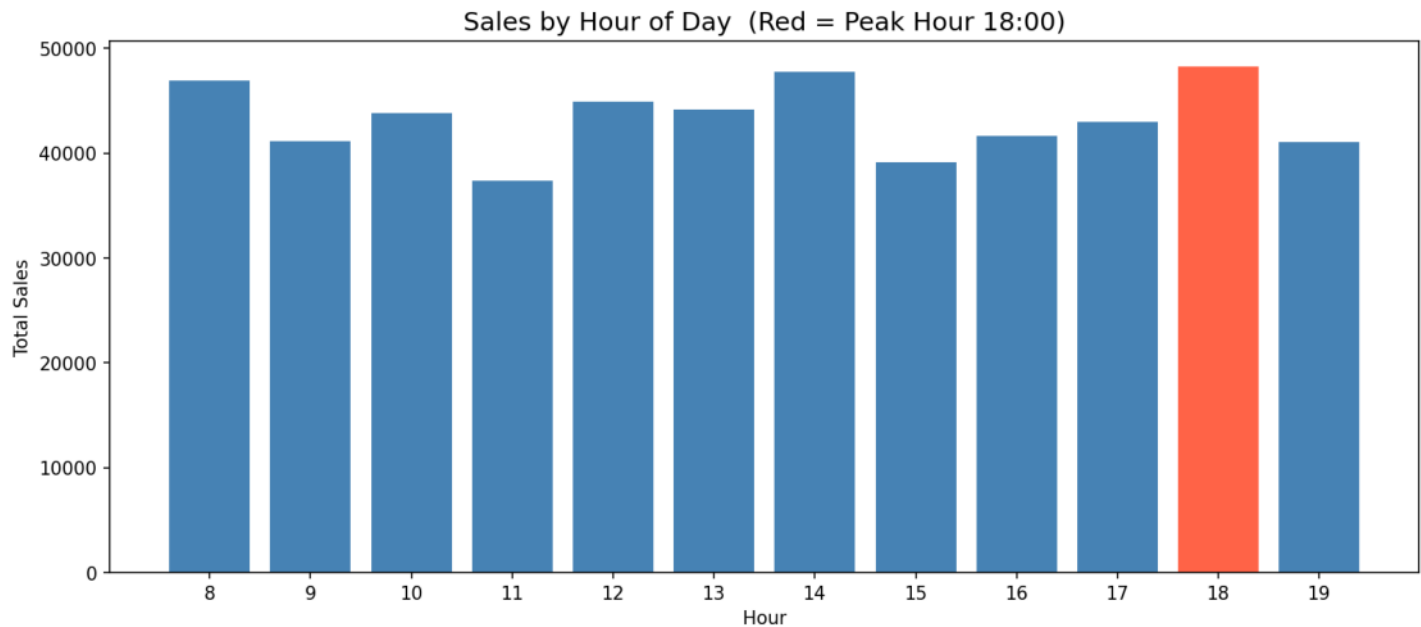


Figure 8

Correlation Between Key Variables

Pearson correlation between numeric variables. Unit Price and Total are moderately correlated — higher priced items drive larger totals. Quantity also contributes positively. Tax is perfectly correlated with Total as it is a fixed percentage of every transaction. Rating shows near-zero correlation with spending, confirming satisfaction is independent of transaction value.

Unit Price-Total r	Quantity-Total r	Rating-Total r
0.005	-0.012	-0.009

